Intégrale n° 74 — Résolution

Énoncé. Calculer $I = \displaystyle\int_0^1 \frac{x^2}{(1+x^2)^2}\,\mathrm{d}x$.

On pose $x = \tan t$, $t \in [0, \frac{\pi}{4}]$. Alors $\mathrm{d}x = \dfrac{\mathrm{d}t}{\cos^2 t}$, $\dfrac{1}{(1+x^2)^2} = \cos^4 t$ et $x^2 = \dfrac{\sin^2 t}{\cos^2 t}$:

$$I = \int_0^{\pi/4} \frac{\sin^2 t}{\cos^2 t}\cos^4 t\,\frac{\mathrm{d}t}{\cos^2 t} = \int_0^{\pi/4} \sin^2 t\,\mathrm{d}t = \frac{1}{2}\Big[t - \frac{\sin 2t}{2}\Big]_0^{\pi/4} = \frac{1}{2}\Big(\frac{\pi}{4} - \frac{1}{2}\Big).$$

$$\boxed{I = \frac{\pi}{8} - \frac{1}{4}}$$